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Strategic Budget Optimization through Multi-Criteria Decision Tools. A Case Study of Capex Optimization Exercise through Peak Expansion without Impacting HSE, Asset Integrity or Strategic Business Objectives

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Abstract

This paper presents a systematic methodology developed by a joint taskforce between ADNOC Onshore and its Shareholders to optimize capital expenditure in a challenging expansion business plan environment (+25% production capacity increase over 5 years). The approach ensures a strict guideline approach from the Management with no compromise on the HSE, asset integrity, or any of the strategic business objectives of the JV. Two complementary tools namely "The Bucketing Exercise" and The Capital Efficiency Exercise" were used to identify and prioritize cost optimization opportunities over the 5-years business plan period. The methodology yielded a combined 15% CAPEX reduction for the first year of the business plan while maintaining the production growth targets.

Introduction

In the context of ADNOC Onshore's ambitious expansion strategy (+25% production capacity increase over 5 years), the need for a robust and transparent budget optimization framework became a critical requirement to rationalize and prioritize the capital expenditures over the business plan duration, which increased from approximately by \$4-5 billion per year to more than \$10 billion per year. Accordingly, a Business Plan Review Taskforce Team was established, comprising employees from ADNOC Onshore and each of its Shareholders, as an advisory body to the Executive Leadership Team of ADNOC Onshore. The role of the Taskforce was to review ADNOC Onshore's 2025-2029 Business Plan, to confirm its robustness and identify opportunities and efficiencies to support ADNOC Onshore's optimization of its 2025 Capital Expenditure ("CAPEX"), without compromising the Company's HSE, Asset Integrity, and production sustain and growth mandates. The CAPEX for years beyond 2025 shall also be considered within the Taskforce's mandate, with initial priority given to that for the year 2025. Furthermore, the responsibilities of the joint taskforce have been identified and approved by the management in a formal Terms of Reference document "TOR" as follow:

- a. Conduct a series of comprehensive due diligence review exercises, to provide confidence in respect of key business plan CAPEX drivers, ensuring delivery of all Shareholder mandates against the approved CAPEX five-year business plan 2025-2029 budget ([Reference 1](#)) with an emphasis on 2025 budget for optimization.
- b. Evaluate and identify potential efficiency improvements areas, analyzing aspects such as operational scope of requirements, fields and reservoir level unit technical cost, work schedule prioritization, phasing and/or deferment, and Drilling/Projects efficiency while ensuring no impact on HSE, Asset Integrity, and production sustain and build-up mandates.
- c. Conduct thorough analysis and working sessions to challenge planned expenditure, engaging with key stakeholders to drive improvements and capitalize on other taskforce's outcomes and recommendations (for example, Well Delivery, Inactive Strings, FSOPR & Drilling Efficiency Taskforces) to accelerate the review, as well as addressing Shareholders comments from different technical and financial governance committees.
- d. Compile recommendations and formulate a proposal to ADNOC Onshore's Executive Leadership Team and Shareholders for review and feedback, where update to be provided to the Management on bi-weekly basis.

With over 1,200 budget line items and multiple stakeholders environment, the challenge was to identify cost-saving opportunities without compromising on the Company mandates, making sure if each key planned activity should be kept unchanged, optimized/challenged in terms of scope or timing, or stopped depending on its business priority and CAPEX ([Figure1](#)).

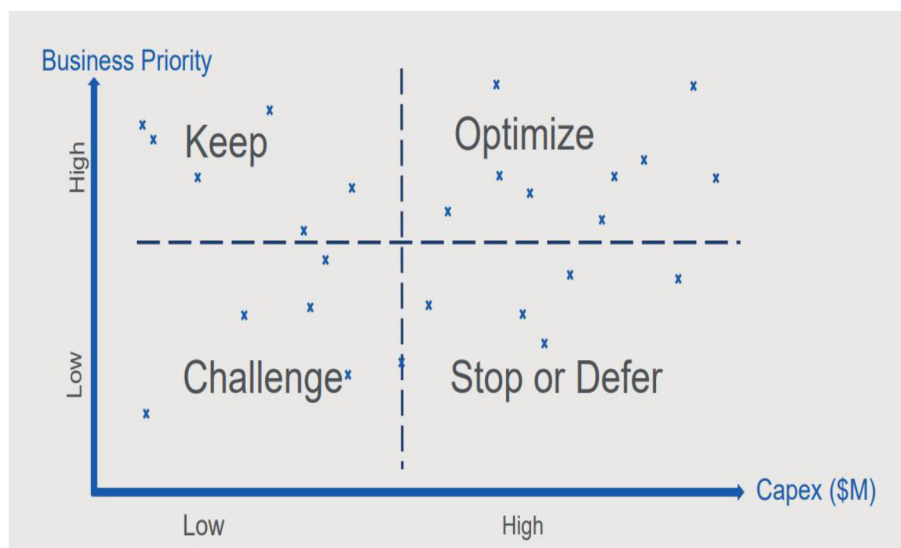


Figure 1—An illustration of CAPEX Optimization Principles

This paper outlines the development and application of a dual-framework methodology that enabled ADNOC Onshore to meet these objectives. The increasing complexity of Oil and Gas operations, coupled with fluctuating market and geopolitical conditions, necessitates innovative approaches to capital investment planning and optimization. This study contributes to the body of knowledge by presenting a replicable model for strategic budget optimization in the petroleum industry which also can be enhanced and expanded into different industries.

Methodology

Taskforce Formation

The formation of the integrated taskforce was a foundational step in ensuring the success of the CAPEX optimization initiative. This process was strategically designed and executed in two parallel streams; internal and external; each contributing to the establishment of a robust, cross-functional, and accountable team capable of driving the optimization agenda forward with precision and alignment. On the internal front, the process commenced with a formal communication from ADNOC Onshore's Executive Leadership to all Senior Vice Presidents (SVPs) across the organization. This communication outlined the strategic importance of the CAPEX optimization initiative and requested the nomination of two focal points from each unit or function. These focal points were selected based on their technical expertise, financial acumen, and familiarity with the operational and strategic priorities of their respective domains. The objective was to ensure that each unit was adequately represented in the taskforce and that the team collectively possessed a comprehensive understanding of ADNOC Onshore's capital investment landscape. The nominated individuals were expected to serve as the single point of contact for their units, facilitating seamless coordination, data gathering, and decision-making throughout the optimization process.

Simultaneously, the external stream of the taskforce formation was initiated through direct engagement with all the seven ADNOC Onshore's Shareholders. A formal request was issued to each of the seven Shareholder entity, inviting them to nominate two representatives to join the taskforce. These representatives were carefully selected to serve as single points of accountability, equipped with the requisite experience, technical knowledge, and financial insight into ADNOC Onshore's operations. The selection criteria emphasized familiarity with both the upstream oil and gas domain and the specific strategic and financial frameworks governing ADNOC Onshore's capital planning. The goal was to ensure that the Shareholder representatives could contribute meaningfully to the taskforce's deliberations and align their respective organizations with the outcomes of the optimization exercise.

Upon confirmation of the internal and external nominations, the taskforce was formally assembled within a one-week timeframe. This rapid mobilization was critical step to maintaining momentum and ensuring timely execution of the optimization activities. A comprehensive kick-off meeting was convened, bringing together the Senior Executive Teams from both ADNOC Onshore and all Shareholder entities. The purpose of this meeting was to align all stakeholders on the taskforce's terms of reference, including its mandate, scope of work, timeframe and expected deliverables.

During the kick-off session, the leadership teams collectively reviewed and endorsed the taskforce's operating envelope, which defined the boundaries within which the team would operate. The session also established a clear feedback timeline, outlining the cadence of progress reviews, interim reporting, and final recommendations.

Furthermore, the workflow of activities was presented and validated, detailing the sequential steps from data collection and analysis to scenario development, stakeholder alignment, and final endorsement. The kick-off meeting concluded with a reaffirmation of commitment from all parties to support the taskforce's objectives and to ensure that the outcomes would be integrated into ADNOC Onshore's planning systems and reflected in the approved budget and to report back to Senior Management with a positive feedback.

This structured and inclusive approach to taskforce formation laid the groundwork for a transparent, collaborative, and results-oriented optimization process. It ensured that all relevant perspectives were represented, that accountability was clearly defined, and that the taskforce operated with the full backing of both ADNOC Onshore and its Shareholders.

Data-Room Establishment

To streamline the sharing of information and establish a solid baseline and reference framework across all internal and external stakeholders, ADNOC Onshore initiated the creation of a comprehensive virtual

data-room. This digital repository was designed to serve as a centralized, secure, and continuously updated platform that enabled instant access to all relevant materials for taskforce members and Senior Management. The data room played a pivotal role in ensuring transparency, consistency, and agility throughout the CAPEX optimization initiative.

The virtual data-room was populated with the latest versions of the 2025–2029 business plan data (Reference 1), detailed budget breakdowns, technical worksheets, and feedback inputs from internal teams. It was structured to support real-time collaboration and facilitate informed decision-making across the integrated taskforce. By providing a single source of truth, the data-room eliminated discrepancies and ensured that all stakeholders operated from a unified and validated dataset.

The cornerstone of the data-room was the development of a single, dynamic pivot worksheet. This worksheet was meticulously constructed using all budget line items from the 2025–2029 business plan (Reference 1) as its foundational dataset. It was designed to be both comprehensive and interactive, enabling users to filter, sort, and analyze data across multiple dimensions. Several categorization and filtering columns were added to enrich the dataset and provide deeper insights into each budget line item. These columns included, but were not limited to (Reference 2):

- Commitment and Project Progress Status: Indicating whether the item was sanctioned, in execution, or pending approval.
- Production Gain/Build-Up Potential: Highlighting the expected contribution of each item to ADNOC Onshore's production targets.
- Budget Evolution Across Review Stages: Tracking changes in budget allocations from initial planning through successive review cycles, including adjustments post-any governance meetings.
- Associated Risk Tier Ranking: Classifying items based on their operational, financial, and strategic risk exposure.
- Bucketing Categorization: Assigning each item to one of the strategic buckets (License to Operate, Business Objectives, Non-Critical), which will be elaborated in the following section.
- Potential Optimization/Deferral Value (Hopper): Estimating the cost-saving or deferment potential of each item based on its criticality and timing.

The pivot worksheet was not only a tool for analysis but also a mechanism for alignment. It enabled the taskforce to conduct targeted reviews, identify high-impact opportunities, and support prioritization decisions with data-backed rationale. The worksheet's structure allowed for seamless integration of feedback from various units and facilitated cross-functional collaboration as a single point of truth.

Moreover, the data-room was configured with access controls and version tracking to ensure data integrity and confidentiality. Taskforce members could upload supporting documents, annotate budget items, and flag issues for escalation. The platform also supported dashboards and visual summaries to aid executive-level reviews and presentations.

In essence, the establishment of the virtual data-room and its embedded pivot worksheet was a transformative step in ADNOC Onshore's CAPEX optimization journey. It provided the infrastructure necessary for disciplined planning, transparent governance, and strategic foresight, reinforcing the organization's commitment to operational excellence and shareholder value creation.

Bucketing Exercise

The Bucketing Exercise involved a systematic categorization of the Company portfolio of activities and associated budget line items into three distinct buckets: 1) License to Operate, 2) Business Objectives, and 3) Non-Critical (Figure-2). First, Bucket #1 includes all budget line items related to laws, regulations, HSE, assets integrity, and business continuity aspects of the business. Second, Bucket #2 includes all budget line items related to annual production mandates, concession obligations (e.g. production and ultimate recovery per reservoir units), digital & automation, and other strategic objectives such as the reductions of

Greenhouse Gases (GHG) emissions towards net-zero, the increase of enhanced oil recovery mechanisms, the appraisal and de-risking activities of complex and tight reservoirs, etc. Finally, Bucket #3 includes all budget line items related to non-critical, low priority or low value activities and which was not part of Bucket#1 or Bucket#2 categorization.

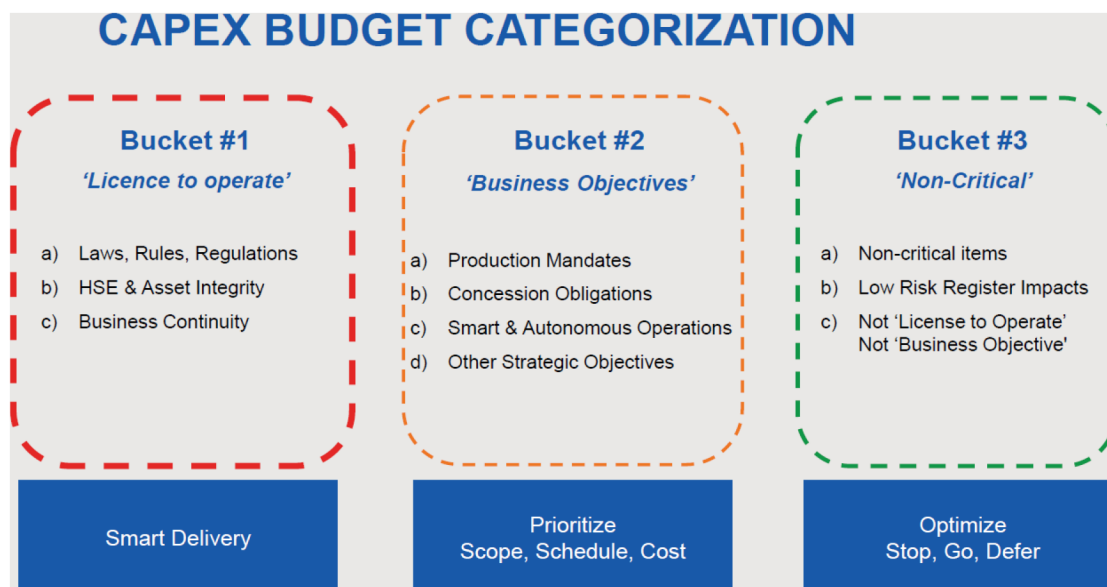


Figure 2—Description of Bucketing Exercise Categorization and Scope

The taskforce challenged each Buckets separately, starting from Bucket #3 to identify quick-wins efficiency optimization or deferment. The next step followed was to review Bucket #2 to challenge the scope of work, priority level, execution schedule and unit cost of each planned activity (studies, projects, wells, equipment, ...). The final step was challenging Bucket #1 aiming for smart delivery of all activities needed to ensure the Company's license to operate and business continuity, i.e. challenging the risk level, urgency and minimum design requirement of each line item.

The detailed Bucketing analysis focused on individual budget items above \$5 million representing approximately 20% of the line items (+/- 200 items) but approximately 90% of the overall CAPEX budget. For the remaining 80% line items below \$5 million (+/- 1000 items) representing only 10% of the overall CAPEX budget, a top-down cost reduction management overview (MOV) target of 15% was given to the various budget owners for the first budget year, giving them the flexibility and time to review and arbitrate within their entity to reach the overall target before final budget approval.

This bucketing classification enabled the identification of optimization opportunities across a massive budget landscape. Each line item was evaluated based on its criticality to operations, regulatory compliance, and strategic alignment, while making sure also that the scope of work, cost and execution timeline were fit-for-purpose and not too conservative. Furthermore, the process included intensive workshops with key organization stakeholders and subject matter experts, historical data analysis, and risk assessments to ensure accurate categorization. The categorization also helped in aligning departmental priorities and facilitated cross-functional collaboration, alignment and consistency in budget planning.

Keeping in mind that Bucket-2 represents 90% of the budget line items, a sub-categorization by Tiers was introduced to help prioritize the various strategic objectives compared to their added-value and business criticality (Figure-3).



Figure 3—Description Sub-Categorization by Tiers for items under Bucket#2

Additional details were gathered for each budget line-items above \$1 million to improve the quality and output of the Bucketing Exercise such as its criticality level (high-medium-low) and its maturity level (preliminary, conceptual or defined study level, sanctioned or not, committed or not, in execution or not, % progress of execution).

Each budget line item subject to cost optimization or deferral was tagged in a "hopper of opportunities" for final review and validation by the management before being captured by Finance as a firm cost reduction actual number.

Capital Efficiency Exercise

The Capital Efficiency metric was introduced as a supplementary tool to rank projects in terms of production delivered per CAPEX invested (see Figure-4). Projects were evaluated in terms of their cumulative capital efficiency over time (from 2025-2040) and prioritized based on their ability to deliver early production gains with minimal risks. The taskforce further developed a decision matrix that incorporated technical feasibility, economic impact, and strategic fit. Scenario analysis and sensitivity testing were conducted to validate recommendations and ensure robustness under varying market conditions. This exercise also included benchmarking between different fields and assets and historical performance data to ensure realistic projections.

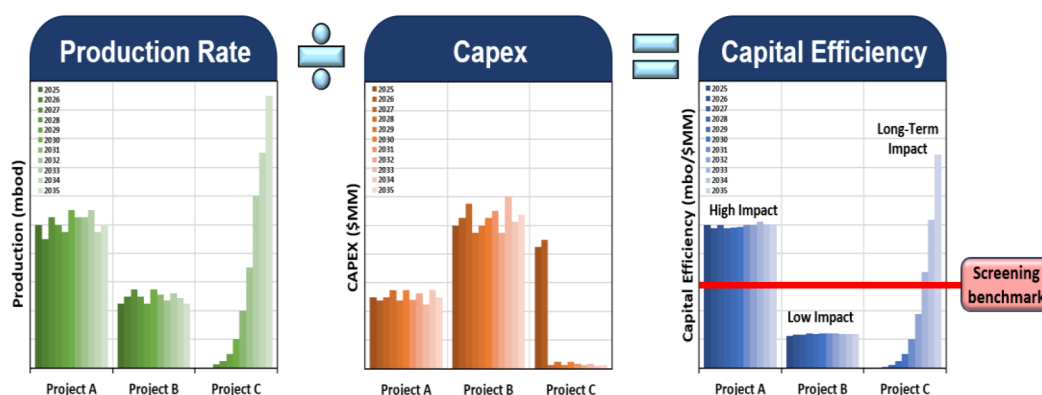


Figure 4—Illustration of Capital Efficiency Metric Formula

All the key capex projects were then ranked in terms of their contribution to the annual oil production vs. their CAPEX intensity to identify the most efficient projects (Figure 5). Perhaps unsurprisingly, base maintenance-related projects delivered the highest value and most immediate production returns. Some projects (such as artificial lift upgradation) delivered extremely high returns in the long term, justifying their prioritization in the business plan despite relatively low returns in the short-term. Other projects (such as enhanced oil recovery) were found to be challenged across the entire duration of the business plan, which flagged them for further scrutiny and review with the asset management teams.

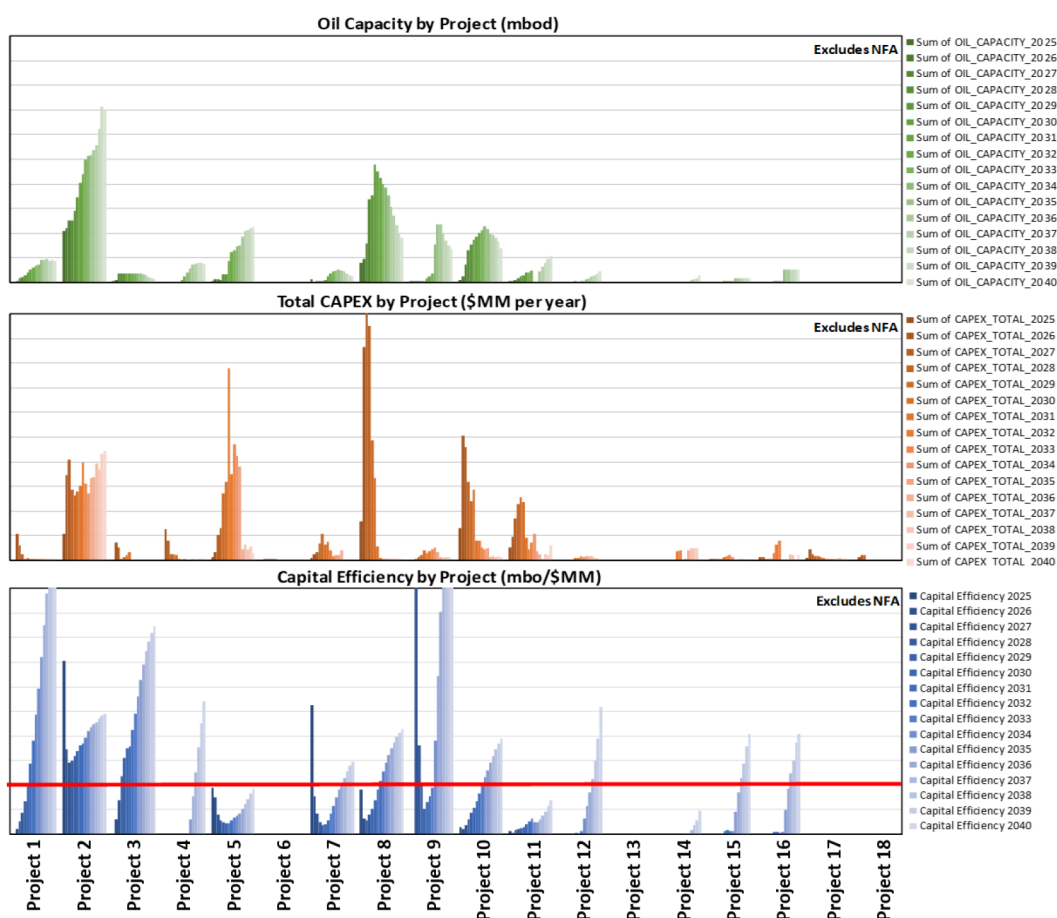


Figure 5—Capital Efficiency Ranking

The outcome of the Capital Efficiency screening process was a funnel of opportunities presented to Management for review and arbitration (Figure 6). The possibility to compensate for the deferred production of low value/high-cost building blocks with accelerated production from high value/low-cost building blocks was also discussed.

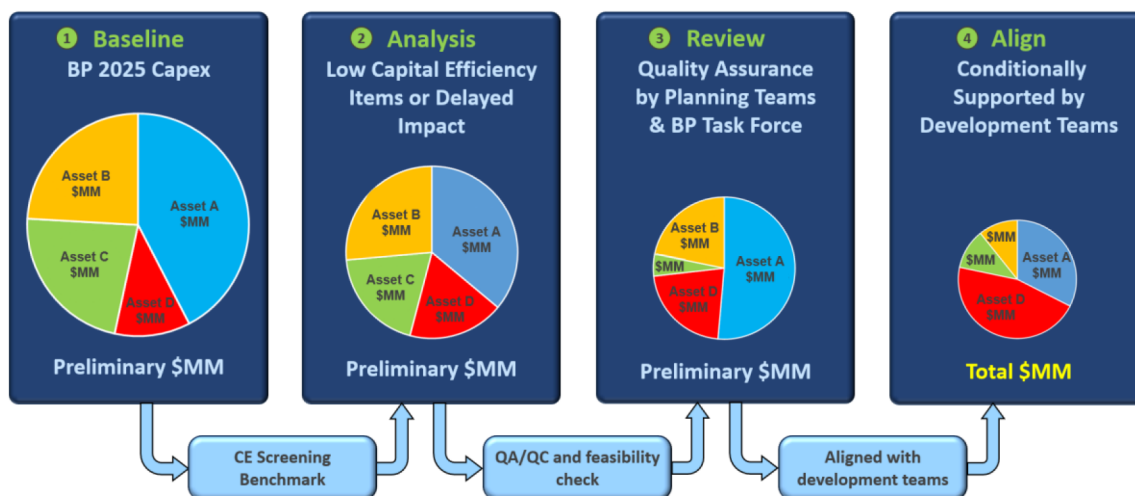


Figure 6—Capital Efficiency Screening Process

Drilling representing approximately 30% of the capital expenditures, the requirement for new wells and workovers was also challenged depending on their priority level (e.g. well integrity, production baseline, production growth, production plateau, de-risking of complex reservoirs, EOR projects pilots, exploration of new reservoirs, ...) making sure that the high value and low cost/risk wells were prioritized over the low value and high cost/risk wells (Figure 7), and that all wells had a firm tie-in plan to put them into production or injection and valorize them as soon as possible.

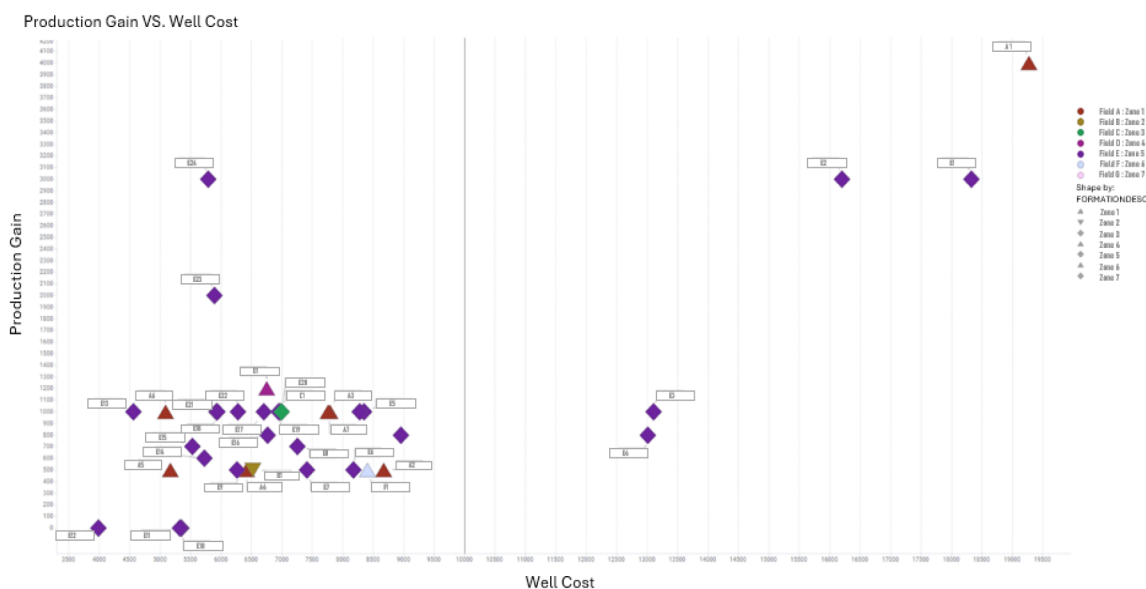


Figure 7—Technical Rate vs Well Cost

Process Workflow

The Process Workflow for ADNOC Onshore's CAPEX optimization implementation begins with a series of extensive and challenging sessions focused on key operational and strategic topics. These sessions

cover areas such as the well delivery process, wells tie-in plan, extended reach wells, inactive strings, well digitalization, Enhanced Oil Recovery (EOR) plans, undeveloped fields and reservoirs plans, exploration wells, committed line items, HSE and Asset Integrity line items, drilling budget optimization, and wells downhole completions.

During these sessions, stakeholders rigorously evaluate each topic to identify potential optimization opportunities. The insights and recommendations from these sessions are then consolidated and transitioned into a second phase of alignment. This phase involves engagement with budget holders and leadership management to review, endorse, and validate the proposed optimizations.

Following endorsement, the workflow progresses to the executive leadership team for final alignment and approval. Furthermore, budget optimization status update is being shared in a timely manner with all Shareholders to ensure a seamless update on budget status during the business plan development proves. Once approved, the optimized budget and strategic decisions are integrated into the planning system, ensuring that all changes are reflected and actionable within the organizational framework.

This structured workflow ensures that CAPEX decisions are data-driven, strategically aligned, and supported across all levels of the organization. It promotes transparency, accountability, and continuous improvement in budget planning and execution.

Results and Discussion

Bucketing Outcomes

The Bucketing Exercise identified approximately 6% in terms of budget optimization. These opportunities were incorporated into the revised budget with no impact on HSE, asset integrity, or strategic objectives. The categorization process revealed redundancies in non-critical expenditures and enabled reallocation of funds to high-priority areas. Stakeholder feedback confirmed the effectiveness of the approach in aligning budget decisions with operational mandates. The exercise also improved transparency and accountability in budget allocation, fostering a culture of fiscal discipline.

Capital Efficiency Outcomes

An additional 9% in optimization was achieved through the Capital Efficiency Exercise. This approach enabled the prioritization of projects with high production value and minimal Capex and risks, accelerating delivery and enhancing shareholder value. The decision required extreme careful alignment assurance measures with internal stakeholders to ensure no impact on future production ramp up and building blocks to avoid impacting future mandates and concession obligations. The decision matrix facilitated transparent discussions among stakeholders and supported data-driven investment decisions. The exercise also highlighted opportunities for portfolio rationalization and phased implementation of capital projects. It provided a clear roadmap for investment planning in case of further capital expenditure constraints due to market or geopolitical conditions for Shareholders to support building inform decision making process.

Strategic Impact

The methodology facilitated a comprehensive review of subsurface and surface requirements under a unified framework. It supports decision-making across a complex stakeholder environment and is now embedded into future business planning cycles. The integration of financial and operational metrics ensured alignment with corporate objectives and enhanced accountability. The approach has been recognized as the best practice within ADNOC Onshore. It also laid the foundation for continuous improvement initiatives and strategic foresight in capital investment planning and setting up a platform of communication across all stakeholders in an increasing cost environment.

Conclusion

The dual-framework methodology proved to be a highly effective tool for budget optimization in a budget constrained environment. It enabled ADNOC Onshore to maintain its expansion trajectory and competitiveness in the market while delivering significant cost savings and strategic alignment. The approach is scalable and adaptable, offering a replicable model for other operating companies facing similar challenges. Future enhancements may include integration with real-time analytics platforms and machine learning algorithms to further improve decision-making capabilities. The methodology also supports long-term sustainability goals by promoting efficient resources utilization and strategic investment planning.

Novel/Additive Information

The Bucketing and Capital Efficiency Exercises represent a novel contribution to budget optimization practices in ADNOC Onshore. Their integration into the annual business plan and budget processes demonstrate their value in enhancing decision-making, stakeholder alignment, and investment efficiency. The methodology bridges the gap between financial planning and operational execution, providing a holistic view of capital allocation. It also fosters a culture of continuous improvement and strategic foresight within the organization. The tools developed are now being considered for integration into digital planning platforms to enhance agility and responsiveness.

References

1. *ADNOC Onshore 2025-2029 Business Plan Document.*
2. *ADNOC Onshore approved CAPEX Optimization joint Shareholders taskforce team Terms of Reference (TOR).*